

Clothes Store Management System – Software Requirements and Implementation Specifications

User Requirements

The Clothing Store is looking for an electronic website system. This system will be used by the customers to browse product selection available with possibility to order a product they want. Also, this system will be used by employees of the store to help them track store information like amount of products available, check orders and track their shift time.

Customers can view products available in the store and potentially place an order of one of them. They can search for their product using a search bar or just find their desired product by just scrolling in a list of products. If a customer wants to order a product, they are required to register their account and present their shipping details. When they are ready, customers are presented with an option of selecting quantity of their product. If the customer does not want to purchase the product now, he is able to create a wishlist and save the product there. Also, he can directly finalize an order. The system must keep track of availability and quantity of given products and inform users in the case that the item is not available. In case that an item is available, but the user's quantity exceeds available stock, customer must be notified about the deficit and ask to lower their quantity of a product in an order. So, customers must be able to modify their order before proceeding to payment.

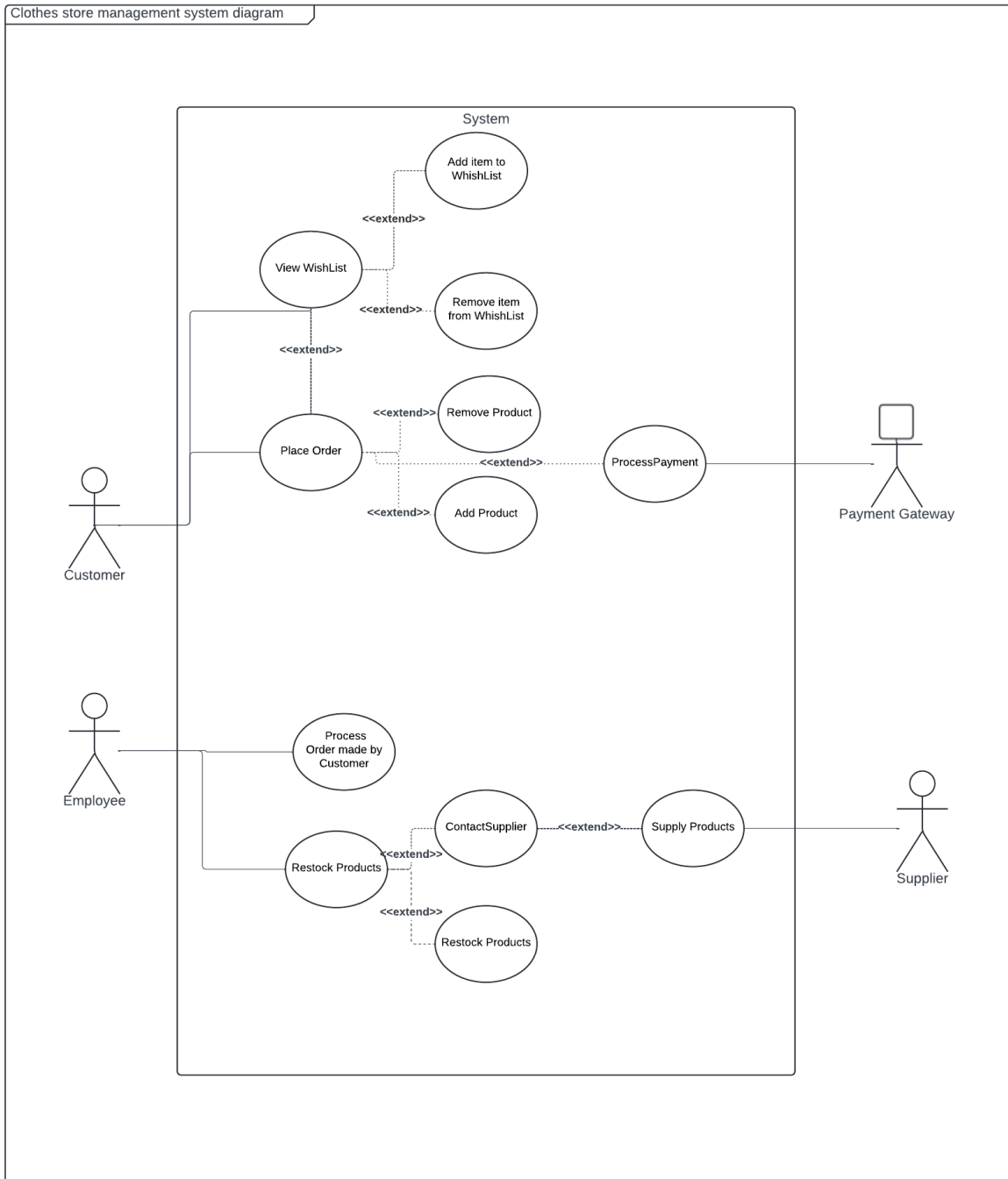
Payments for an order can be done through different payment methods. Transaction must be completed for payment to be finalized. If the transaction is not complete or was abrupted, user has a one-hour period of finalizing the process. Otherwise, the order would be cancelled, and customers will have to create a new order.

Employees through the application can see quantity of products. In the case that the stock will be empty, they will be able to see contact information of a supplier and order more products. They have an option to display their calculated salary and their role. Also, they can see their

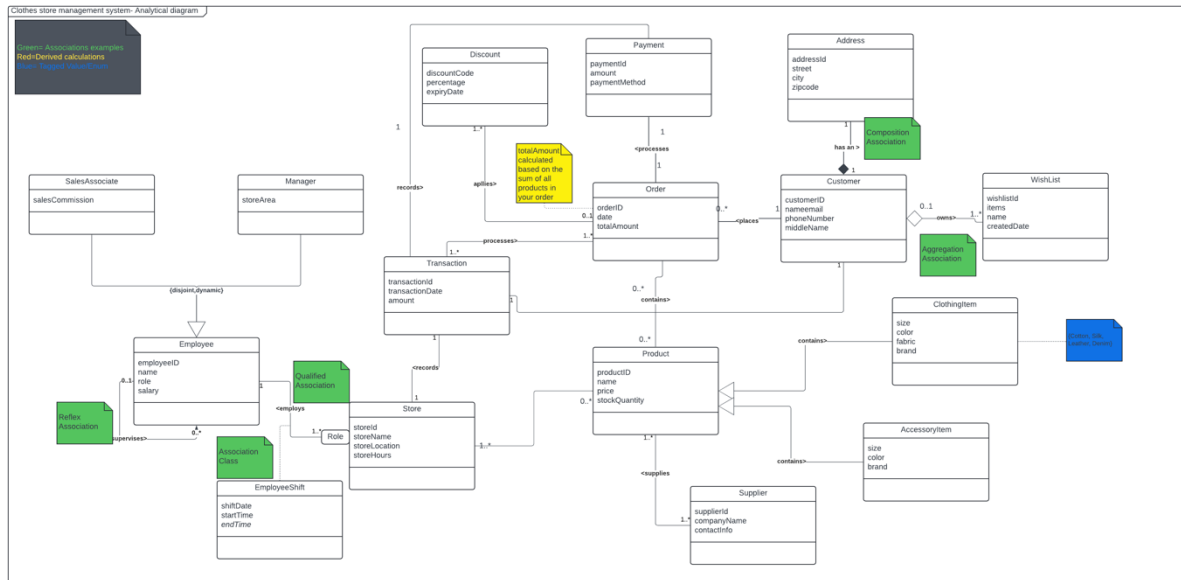
schedule of working hours. They can access their shift history in the past. They can see the information whether they will be supervised by another employee or if they will supervise someone.

Clothing Store System can sell two types of products: clothing items and accessory items. Each of those items contain information about them, like name, price. Also, they have more specific information, for example size, colour, brand etc.

Use Case Diagram



Analytical Class Diagram



Analytical Class Diagram - Description

The following analytical diagram visualizes the Clothes Store Management System where a company can track customer orders, products, manage inventory levels, restocking through suppliers, management of employees and is designed to increase efficiency of the store.

- When customer want to make a purchase in the store, he must have a custom ID, name, e-mail address and a phone number. He also can add an optional middle name. A customer can place one or more orders and can also have wishlist that has all their items saved before proceeding to payment
- Customer can enter its address that can be used when shipping an order. Each address has its street, city, and zip code.

- Customers can create a wishlist before creating an order. A wishlist has its own ID, a list of items added by user, a name, and a created date. When customer proceeds to order, the content of a cart is processed into the order, and the wishlist is being deleted after an order creation.
- When an order is being created by a customer, a custom ID, date of the order, total amount of products is being defined for user that is calculated based on the sum of all products in an order. With a created order, customer then can proceed to payment.
- Payment is created for each order, with payment id, amount that matches the amount in the order and payment method being defined. It can be created only once per order and cannot be used in other orders.
- Each product has their product ID, name, its price and quantity. The quantity can be changed when an order for a product is created.
- Products are supplied by suppliers. Products can be filled in customer's order and one product can be in multiple orders. Also, products are divided into clothing items and accessory items.
- Each clothing item has its own size, colour, brand. It is a specific type of a product and include additional information about a product.
- Another type of product is accessory item that has its size, colour, brand and fabric. Fabric has a set of values (enumeration) that can be assigned to it.
- A store has its own ID, name, and location and opening hours. It employs employees, stores products and accepts transactions.
- Employees have their own ID, name, their role and salary. They are divided into sales associates and managers. Also, one employee can supervise another one. Managers handle store operations, and sales associates work directly with customers.
- Employee shift has its own date, start time and end time and is designed to track worktime for employees.

- Suppliers are a vital part of the store since they provide products to the store. Each supplier has their unique ID, name, and contact information.
- For certain products, discounts can be available, that has a discount code, percentage decreased from a price and expiry date. Discounts can be applied to multiple products or none.

Use Case Scenario 1: “Add an Item to a Wishlist”

Actor: Customer

Purpose and context: a customer wants to save their wanted product to a wishlist so that he would be able to continue with a desired purchase of a product in a later time.

Assumption:

The customer is logged in to their account

The wishlist is already created in the customer's account

Precondition:

The customer has searched for their desired product

Basic flow of events:

1. The customer clicks on the “Add to Wishlist” button on the product page
2. The system confirms that the product was added to their wishlist and prints a success message: “A product was Added to Your Wishlist”
3. The system adds two buttons on the screen and gives two options to the user: to “View an Updated Wishlist” or “Continue Shopping”
4. The customer clicks on “Continue Shopping” button
5. The website displays the main page

Alternative flow of events:

Product already in a wishlist

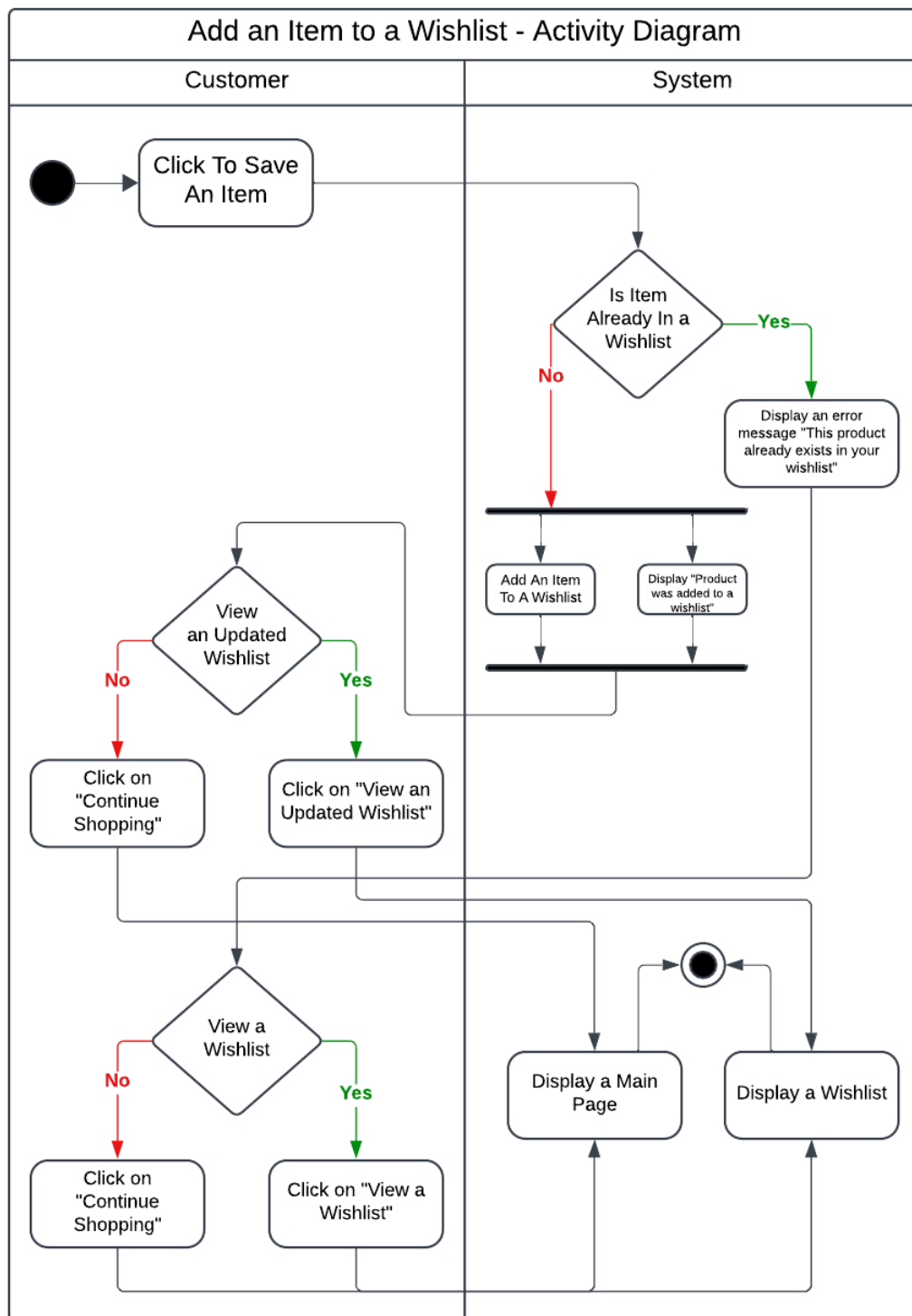
2a1. The system displays an error message: "This product already exists in your wishlist"

2a2. The customer is prompted with the following options: "View a Wishlist" or "Continue Shopping"

Postcondition:

Basic: an item is added to the wishlist

Product already in a wishlist: no duplicate is added, customer is notified



Use case scenario #2 – Placing an order

Actor: Customer

Purpose and Context:

A customer wants to place an order after browsing the product catalog and adding desired items to their cart.

Assumptions:

The customer has already registered and logged into the system. The customer has browsed the product catalog and added items to their cart.

Preconditions:

The customer's account is active, and they have provided valid shipping details. The items they wish to order are in stock.

Basic Flow of Events:

The customer navigates to their shopping cart.

The system displays the list of items in the cart, including their quantity, price, and total cost of the order.

The customer verifies the items and quantity.

The system checks stock availability for the chosen quantity of each item.

The customer proceeds to checkout and selects a payment method (e.g., credit card, PayPal).

The system redirects the customer to the payment gateway and processes the payment. If the payment is successful, the system confirms the order and provides an order summary with the estimated delivery time. The customer can view the order confirmation page or return to the main page.

Alternative Flow of Events:

Item Out of Stock:

4a1. If any item in the cart is unavailable, the system notifies the customer with the message: "Item [item name] is out of stock."

4a2. The customer can either remove the unavailable item or continue with the available items.

Payment Aborted:

4b1. If the payment process is interrupted or unsuccessful, the system notifies the customer with: "Payment not completed. You have one hour to finalize your order before it is canceled."

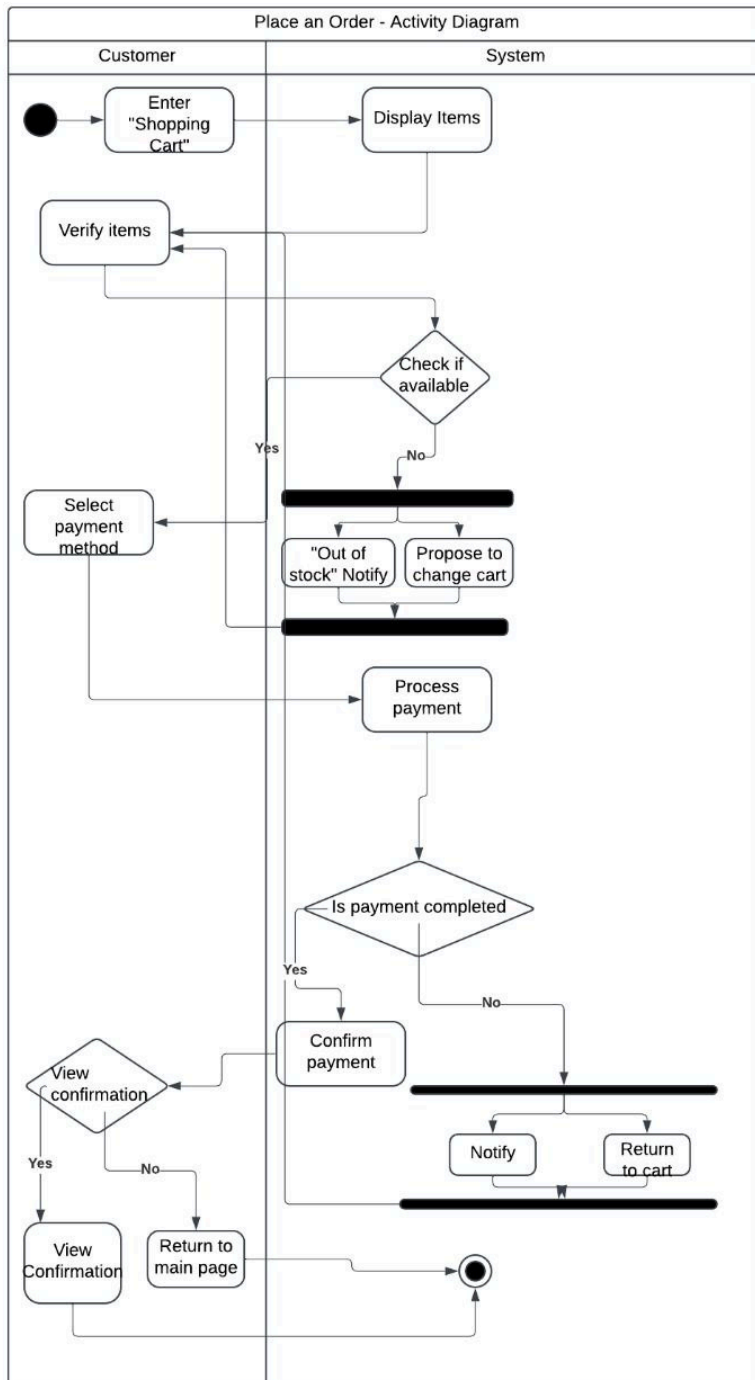
4b2. The customer can retry the payment within one hour or return to their cart to make changes.

Postconditions:

Basic Flow: The order is successfully placed, and the system deducts the item quantity from the store's inventory.

Item Out of Stock: The order cannot proceed until the customer resolves the issue.

Payment Aborted: The order remains pending for one hour, after which it is canceled if no action is taken.



Use Case Scenario #3: "Restock Products"

Actor: Employee

Purpose and Context: An employee needs to restock a product that is running low or out of stock by contacting the supplier.

Assumptions:

1. The product's quantity is monitored by the system, which alerts when the stock is low.
2. The supplier's contact information is stored in the system.

Preconditions:

1. The employee is logged into the system with the necessary permissions.
2. The product in question is marked as low or out of stock in the inventory.

Basic Flow of Events:

1. The employee navigates to the product inventory management page.
2. The system displays a list of products with their stock levels, highlighting those that are low or out of stock.
3. The employee selects the product that needs restocking.
4. The system displays detailed information about the product, including the supplier's contact details.
5. The employee contacts the supplier using the provided information (e.g., phone number or email) to place a restocking request.
6. The employee updates the system with the expected delivery date and quantity of the restocked product.
7. The system updates the inventory status to "restocking in progress."

Alternative Flow of Events:

No Supplier Available:

4a1. If the system does not have supplier information for the product, it displays: "Supplier information not available. Please add supplier details to proceed."

4a2. The employee can add supplier details and retry the restocking process.

Insufficient Restock Quantity:

6a1. If the supplier cannot fulfill the requested quantity, the employee adjusts the restock request.

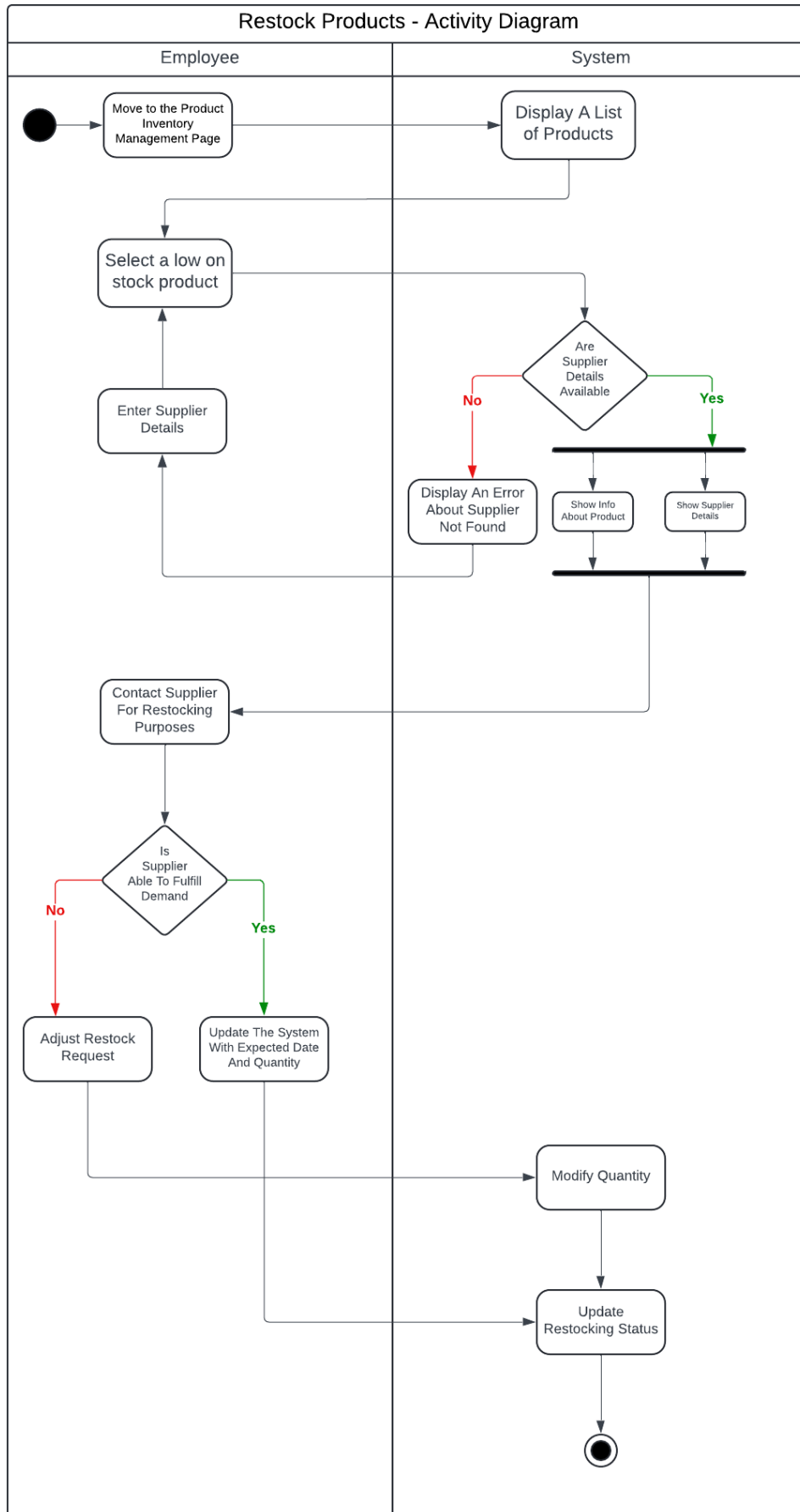
6a2. The system updates the inventory accordingly with the modified quantity and delivery date.

Post conditions:

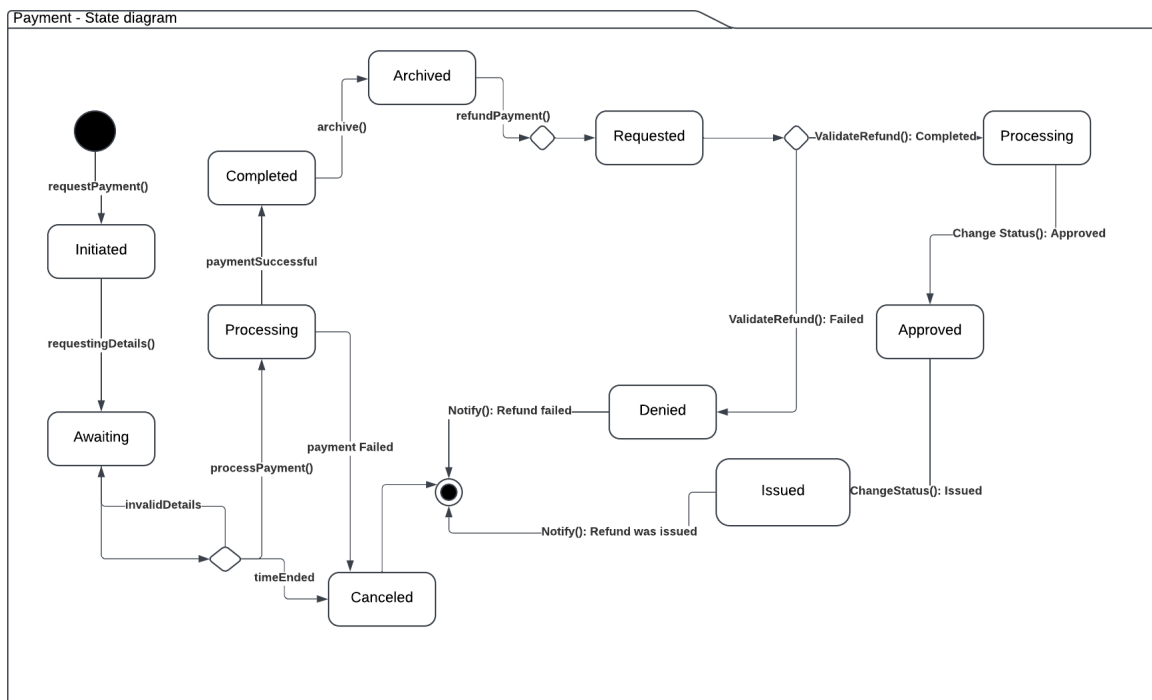
Basic Flow: The product is marked as "restocking in progress," and the inventory will be updated once the restocked items arrive.

No Supplier Available: The restocking process is paused until supplier information is added.

Insufficient Restock Quantity: The product inventory reflects the adjusted restock request.



State Diagram



State Diagram – Design Description

This state diagram describes the states of Payment objects. When using `processPayment()` method, a new Payment object is created with the status of **Initiated**. After requesting payment details, the object enters **Awaiting** state.

Depending on conditions, object can then:

- Enter **Processing** state, if all the details were given correctly
- Repeat the process of waiting for details, if invalid data was received
- Enter **Cancelled** state and then end, if time for giving details has ended

If payment was successful, the object enters **Completed** state. Otherwise, objects enter **the Cancelled** state.

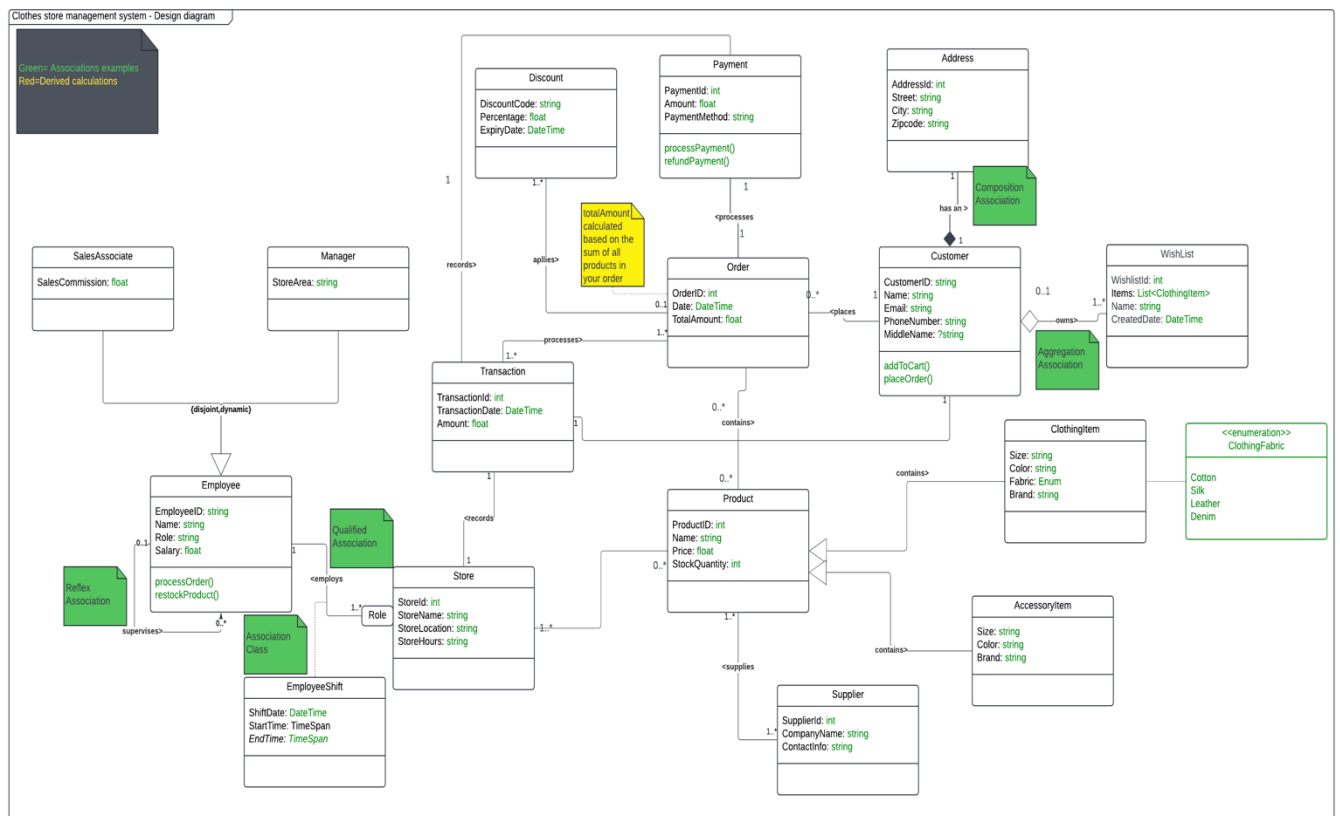
After that, the object becomes **Archived**.

On request, using refundPayment() method, object enters **Requested** state, which then can enter:

- **Denied** state, if refund validation failed, after which user gets notified about the failure, and state ends
- **Processing**, if validation is completed

After being Processed, objects become **Approved and** then **Issued**. User then gets notified about the success of the refund and state ends.

Design diagram



Design Decisions for Clothing Store Management System

The design of the Clothing Store Management System has incorporated several key decisions regarding class modelling, persistence, and attribute handling. These decisions are aligned with best practices in C# and Entity Framework Core (EF Core), ensuring a robust and efficient system architecture. The following sections detail the primary design decisions, including the handling of attributes, associations, and persistence.

Optional Attributes

Optional attributes are modeled using nullable C# properties, marked with ? (e.g., decimal? for nullable decimals). In the database, these properties are represented by columns that allow null values, which are visually denoted in the UML with a multiplicity of [0..1]. For example, MiddleName can be nullable, meaning not every product requires a description.

Multi-value attributes

Multi-value attributes are represented using collections, such as ICollection<T>. These can store lists of values related to an entity. For instance, a Product can have multiple Category associations (e.g., "Clothing", "Winter", etc.), represented as a collection of categories. The system uses EF Core to manage this many-to-many relationship, allowing multiple values to be associated with a single product.

```
public ICollection<Category> Categories { get; set; }
```

Derived Attributes

Derived attributes are properties calculated from other fields. These can be computed using a getter in C# rather than being stored in the database. For example, a Product.TotalPrice could be derived from Price and Discount attributes.

```
public decimal TotalPrice { get => Price - (Price * Discount); }
```

This avoids redundant data storage while providing real-time calculated values when needed.

Tagged value

Tagged values are implemented using enumerations (enums). These are types that represent a set of predefined values. For instance, the ClothingFabric of an order could be an enum, simplifying the representation of possible states.

```
public enum ClothingFabric
{
    Cotton,
    Silk,
    Leather,
    Denim
}
```

Associations

Association is the reference between objects. Each association type has the following methods implemented: adding, removing, and modifying the association.

Basic Associations

For basic associations (1..*) we use the principles of encapsulation for secure storing of a reference.

In an Add method we firstly check on whether the object we want to connect is null. Then we check on whether the connection already exists. If it does not, we assign reference and make a reverse connection.

In the Remove method we firstly check on whether connection is null. If not, then we assign null to the reference and reverse reference.

In the Update method we firstly check on whether the old or new reference is null, and then remove an old reference and add a new one.

Composition

Composition is a whole-part relationship when the part cannot exist without a whole. The main example in the project would be relationship between Address and Customer.

The Remove method differs from the basic association is that it ensures that the reference to the Address in the Customer is cleared, and the Address object is being deleted because it cannot exist without a Customer.

The Set method ensures that an Address cannot be reassigned to a different Customer without clearing the existing relationship.

Aggregation

Aggregation is a “has-a” relationship. In this case depended on object can exist independently, compared to a composition. The main example would be a relationship between Customer and a Wishlist.

Qualified Association

Qualified association is association with a key assigned to it. In the project the main example would be a relationship between a Store and Employee where the Role serves as a key. The main difference between implementation of a basic association would be the reference itself, that is not an object or a list of objects, but a Dictionary<string, Employee>. And based on that Add, Remove, Update methods now modify the dictionary itself.

Association Class

This type of association includes an intermediate class that contains attributes for the relationship between classes. The main example would be EmployeeShift class that contains information about shifts between Store and Employee.

It includes SetEmployee and SetStore classes that add the reference of those objects to EmployeeShift object.

Reflex Association

Reflex association is when objects of the same class are being related to one another. The main example would be an Employee class with an implementation of Subordinates and Supervisor.

It includes the Set and Remove method.

Association Validation

We have used different ways to validate associations:

Null check: all Add and Remove methods have null checks implemented for an empty reference not to be assigned.

Duplicate Prevention: methods ensure that no duplicate references will be added to collections.

Reverse References: Each association ensures that the reverse connections are updated and assigned.

Exception Handling: in case of an error, appropriate exceptions would be thrown.

Dynamic Inheritance

We have implemented a new constructor for the child classes that takes the base class as an argument. Then we have created a SwitchFrom method that creates a new object of the parent class, but from the opposite child class. Then removes the old object from the extent.

Multi-Inheritance

To create a multi-inheritance we have created new interfaces, like IClothing, IProduct to replace one of the inheritances.

Dynamic Analysis on use case scenarios

After further work on the use case scenarios, we have performed a dynamic analysis on them. After it, we saw that further expansion of a design class diagram is needed to fully implement features that are presented above. The new methods are also included in the updated class diagram below.

Add an Item to a Wishlist

New methods:

`showProductInformation()` – this method is used to get the product information details, like Name, Price, StockQuantity, Size, Color, Brand, Fabric from the product and returns them.

`checkProduct()` – this method checks whether the product that user checks has StockQuantity higher than his requested stock.

`displayItems()` – this method get all of the products that are saved in the list of ClothingItems and returns them.

`createWishlist()` – this method is used to create a new Wishlist that is connected to the Customer itself, since the association between classes is aggregation. Customer can enter a name of it.

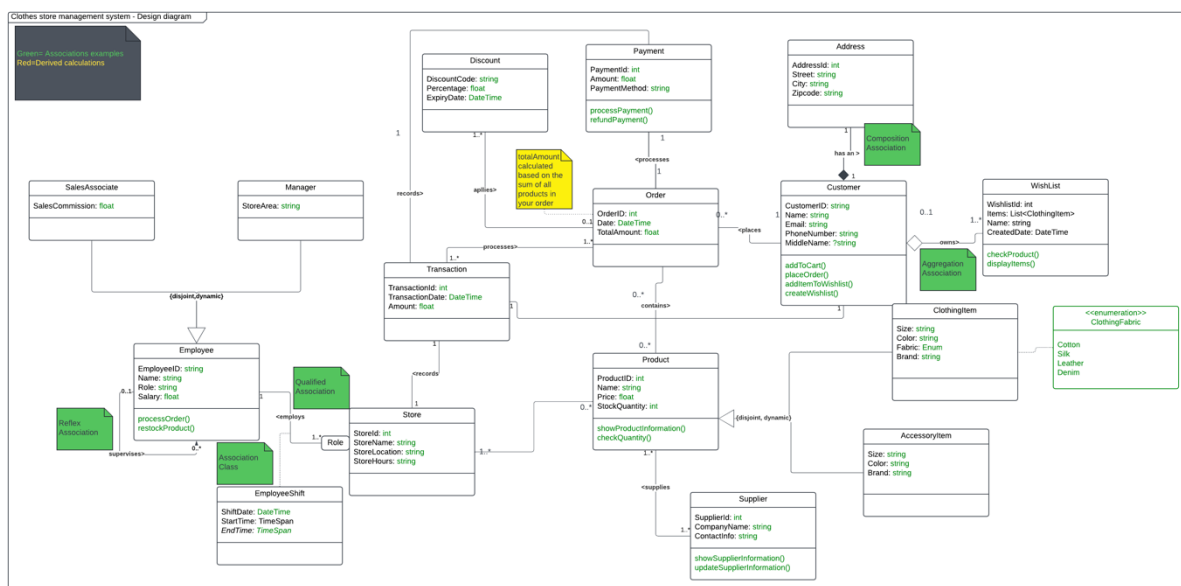
`addItemToWishlist()` – this method is used to insert products into a Wishlist object that is connected to the Customer itself. This method needs a Wishlist to enter Product into and a Product itself.

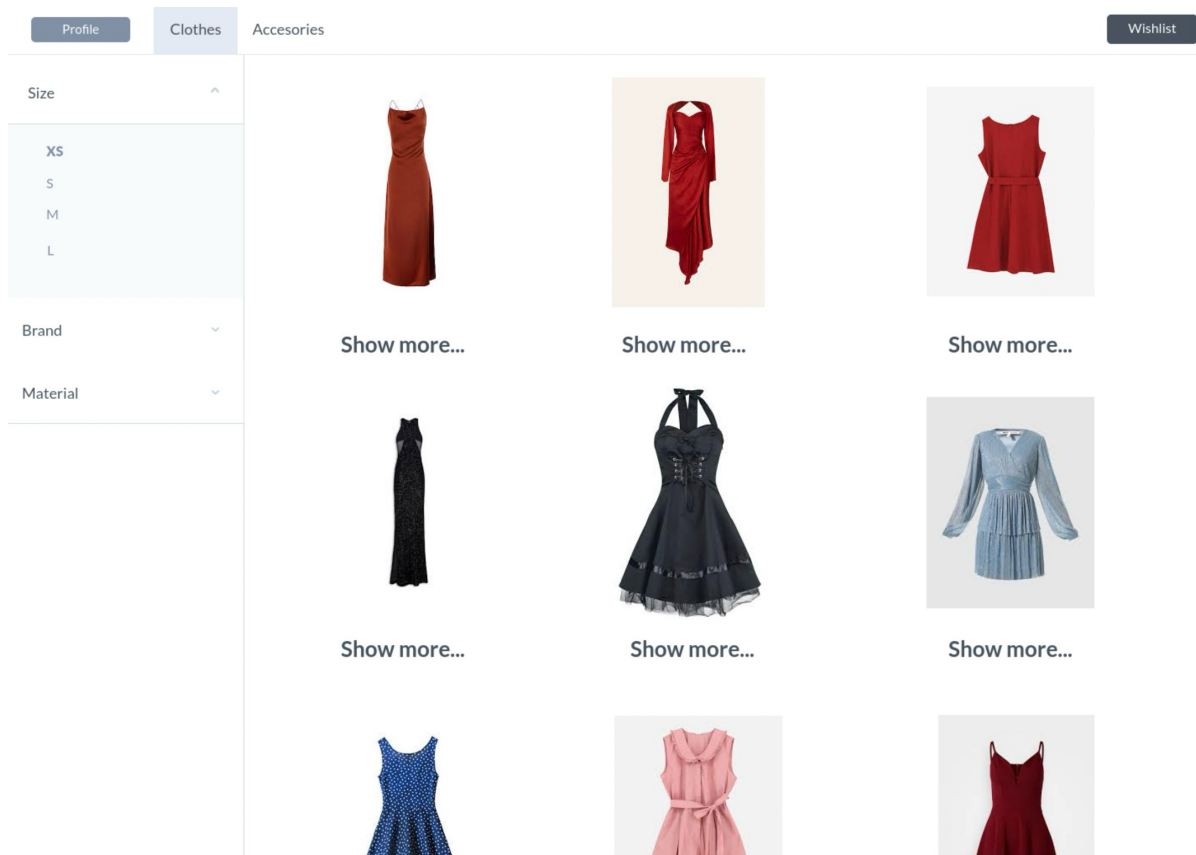
Restock Products

`checkQuantity()` – this method checks whether the StockQuantity number of a product is below five.

`showSupplierInformation()` – this method gets the information of a Supplier object and returns them.

updateSupplierInformation() – this method is used to modify the main information of Supplier, like companyName, contactInfo.





Main page

The screenshot on the top describes the appearance of the main page of our site with existing categories, additionally a wishlist button can be seen that allows the users to access their wishlist page and finalize their order with the selected products.

Placing an order

Profile

Clothes

Accessories

Wishlist



Dress QLI7-RT5E-89OI

M

-

1

+

25,99 \$



Dress QLI7-RT6E-88OU

S

-

1

+

28,99 \$

TOTAL: 54,98 \$

Discount code

Place order

Place an order(Step 1, 2, 3 and 4)

The following page illustrates the displayed info of a given products after the user has browsed the product catalog and added items to their wishlist. When the customer navigates to their wishlist, each item is displayed with its associated details, including quantity, price per unit, and the total cost of the order calculated dynamically. The customer can review and verify the listed items and their quantities, ensuring accuracy before proceeding. In addition, the customer can apply a discount code. For each item, the system performs a real-time check of stock availability to confirm that the requested quantities can be fulfilled.

Profile

Clothes

Accessories

Wishlist

TOTAL: 54,98 \$

Choose payment method ▾

PayPal

Blik

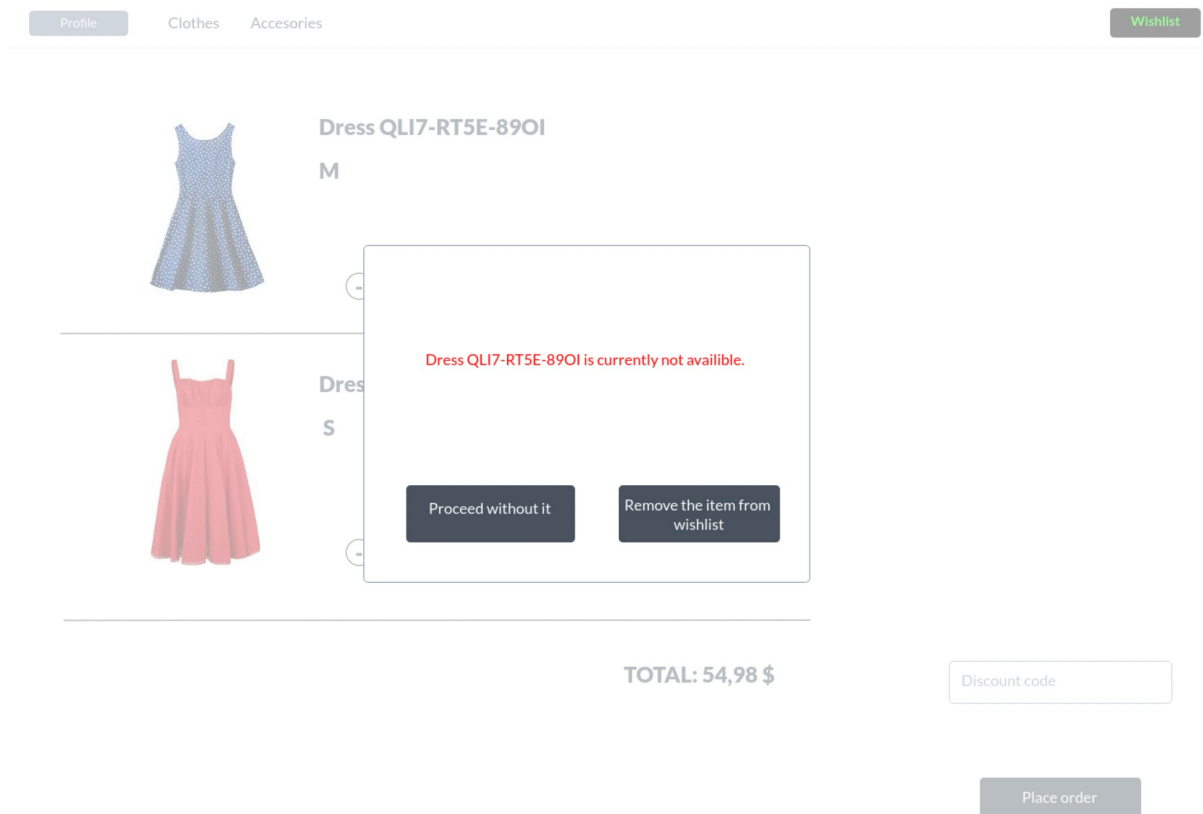
CreditCard

Bitcoin

Pay

Place an order(Step 5,6)

After the customer was convinced of his order and pressed the button to place the order he was transferred to the page of choosing a payment method. The customer proceeds to checkout and selects a payment method (e.g., credit card, PayPal). Then the system redirects the customer to the payment gateway and processes the payment.



Place an order(Step 4a1, 4a2)

System notifies a customer that the product in your wishlist is not unavailable right now. The customer is prompted with an error that "Item [item name] is out of stock." You have two options: The customer can either remove the unavailable item or continue with the available items.

TOTAL: 54,98 \$

Choose payment method ▾

PayPal

Blik

CreditCard

Bitcoin

Payment not completed. You have one hour to
finalize your order before it is canceled

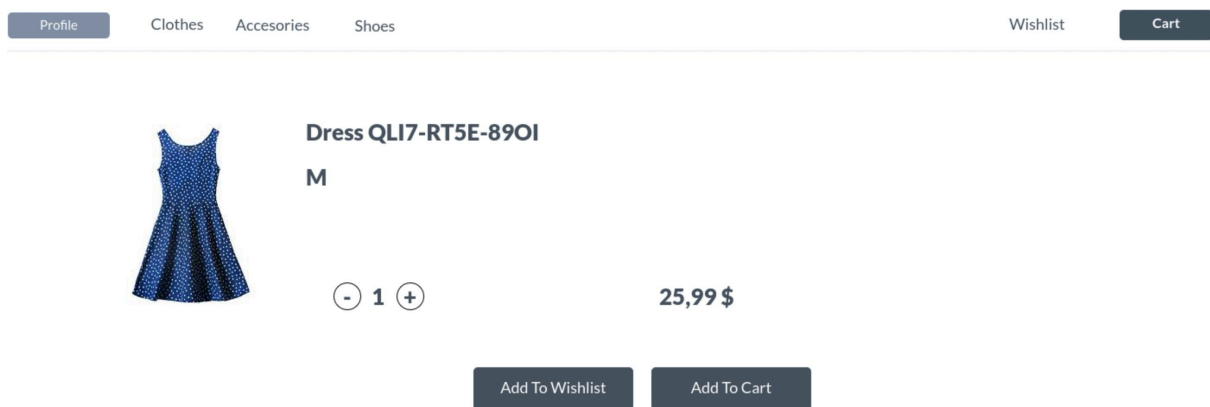
Retry payment

Return to wishlist

Place an order(Step 4b1, 4b2)

System notifies a customer that payment process is interrupted or unsuccessful. The customer is prompted with an error that: “Payment not completed. You have one hour to finalize your order before it is canceled.” You have two options: The customer can retry the payment within one hour or return to their cart to make changes.

Add An Item to a Wishlist



Adding an Item to a Wishlist (Step 1): when choosing desired product, the customer is presented with information about product (name, photo, size, price). Also, user is presented with the option of adding this item to a wishlist.



Dress QLI7-RT5E-89OI

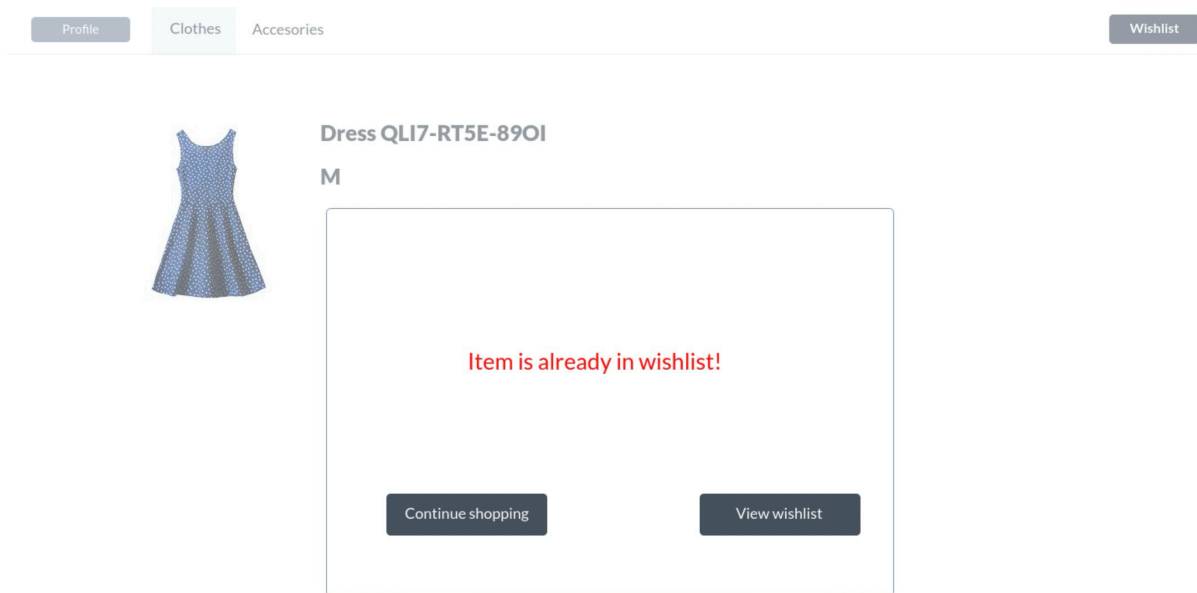
M

Item added to your wishlist!

Continue shopping

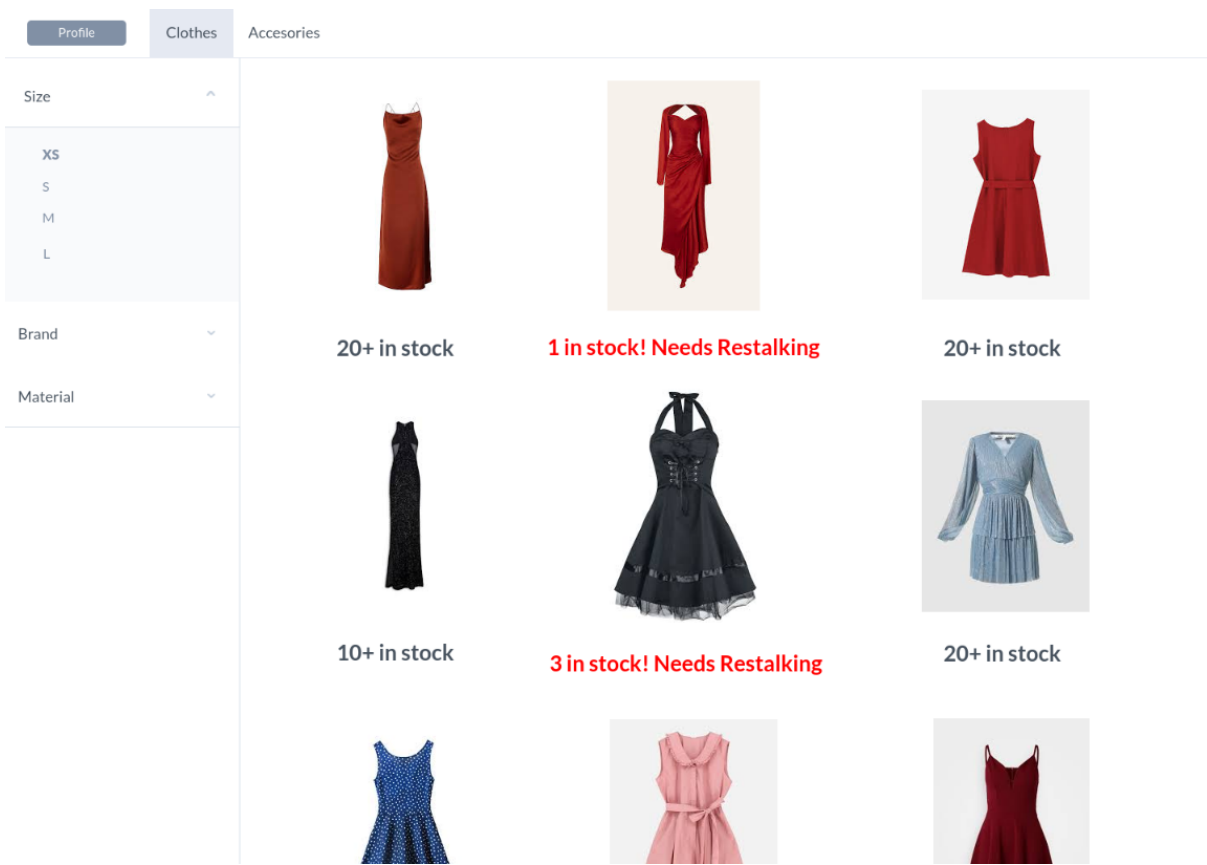
View wishlist

Adding an Item to a Wishlist (Step 2-5): User is prompted that their product was added to a wishlist and is presented with an option of returning to “Continue Shopping” (returning to the main page) or to “View an updated Wishlist”.



Adding an Item to a Wishlist – Alternative: 2a1-2a2 (Product Already in a Wishlist): The customer is prompted with an error that “This product already exists in your wishlist”. And is given two options: “View a Wishlist” or “Continue Shopping”.

Restock Products



Restock Products(Step 1, 2 and 3)

The following page illustrates the process of managing product restocking within the inventory management system. When an employee navigates to the product inventory management page, the system provides a detailed view of all products, highlighting those that require immediate attention due to low or zero stock levels. Employees can then select a specific product to initiate the restocking process, ensuring inventory accuracy and timely replenishment.



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M

25,99 \$

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fronces@gmail.com

+1 111-111-112

Restock



Dress QLI7-RT5E-89OI
M

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+1 111-111-112

Restock

Update the system

< October 2017 >						
Su	Mo	Tu	We	Th	Fr	Sa
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Quantity

Supplier Details

Delivery date

Restock

Restock Products(Step 4, 5 and 6)

When an employee selects a product requiring restocking from the inventory management page, the system displays detailed information about the product, including supplier's contact details. The employee uses the provided supplier contact details to place a restocking request. Communication can be initiated through: phone call or email. The employee specifies the required quantity of the product to the supplier. After confirming the order with the supplier, the employee updates the system with the restocking details, expected delivery date and quantity of the products ordered

Profile

Clothes

Accessories



Dress QLI7-RT5E-89OI

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fronces@gmail.com

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Restock

Update the system

< O

Su	Mo	Tu
1	2	3
8	9	1
15	16	1
22	23	2
29	30	3

Restocking in progress

Delivery date

Restock Products(Step 7)

The system dynamically updates the inventory status for the relevant items, setting it to "restocking in progress." This status change ensures that users and administrators are informed of the ongoing replenishment process and prevents further orders for the affected items until restocking is complete.

Profile

Clothes

Accesories



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+1 111-111-112

Restock

Update the system

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Su	Mo	Tu
1	2	3
8	9	10
15	16	17
22	23	24
29	30	31

Delive

Supplier Information not available. Please add supplier details to proceed

Restock Products – Alternative 4a1.(No Supplier Available): The system displays a prompt to an Employee :“Supplier Information not available. Please add supplier details to proceed”.

Profile

Clothes

Accessories



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Restock

Update the system

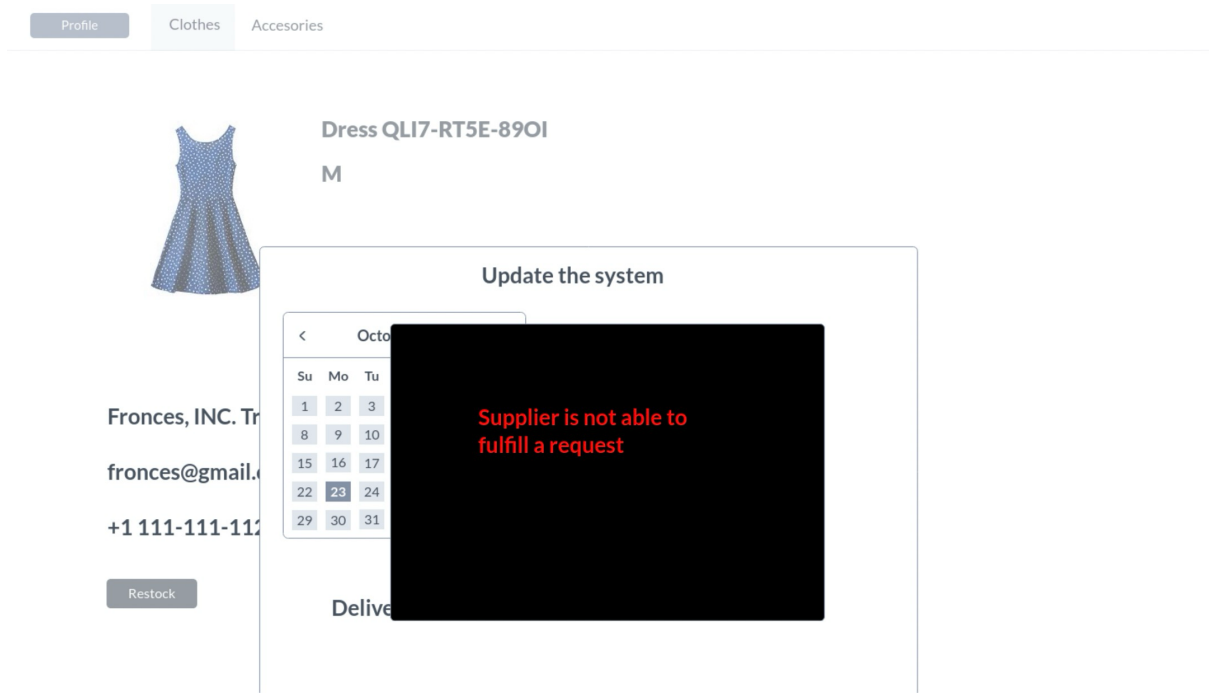
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Su	Mo	Tu
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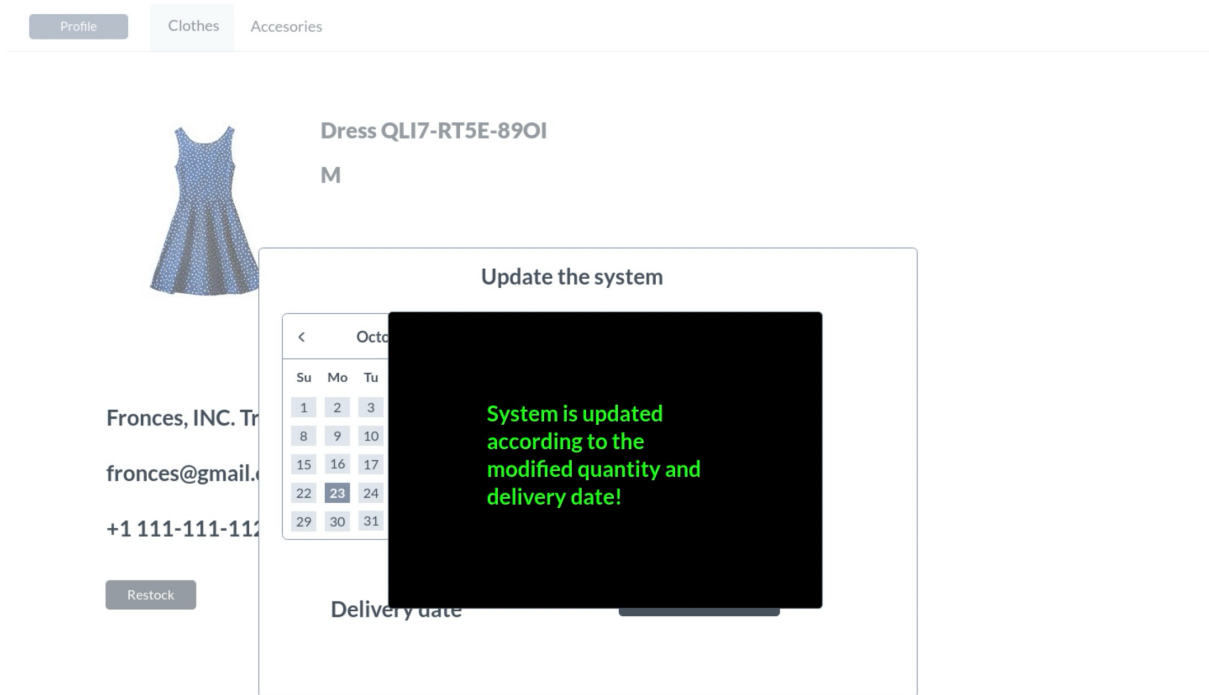
Delive

Supplier Information updated, you may retry restocking

4a2. When supplier information is updated, employee is notified about that, and he is given an option to retry restocking process.



Restock Products – Alternative 6a1 (Insufficient Restock Quantity): Employee is prompted with an error: “Supplier is not able to fulfill a request”, employee is able to adjust restocking request to allowed number.



6a2. The user is prompted that the system is updated according to the modified quantity and delivery date.